

The Real System

The problem is not
food waste, it is
coordination

Surplus Food · System Analysis





Colombia generates 9.7 million tons of food waste annually.

The District Initiative "Yo Contribuyo" was launched in October 2024.



The walkable radius around UD



But the Problem Is Only 3 km Away

Why does surplus food fail to reach those who need it when
they are so close?

Stakeholders

The System Stakeholders



Cafeterias
Donor



Restaurants
Donor



Students
Beneficiaries



Community members
Beneficiaries



Verified organizations
Beneficiaries



Others
Beneficiaries

Real constraints

The Constraints That Define the Game

Temporal

Food expires the same day. Pickup window: 2–4 hours; collection time: 1–3 hours.

Spatial

Maximum walking distance of 3 km with center in UD

The *IPO* model



Inputs: surplus food + geolocation + preferences.



Outputs: redistribution + impact metrics.

“

These are not software requirements. They are properties of the social system.

The 10 System Sensitivities

With an example

Critical Inputs

2

No-shows

75%

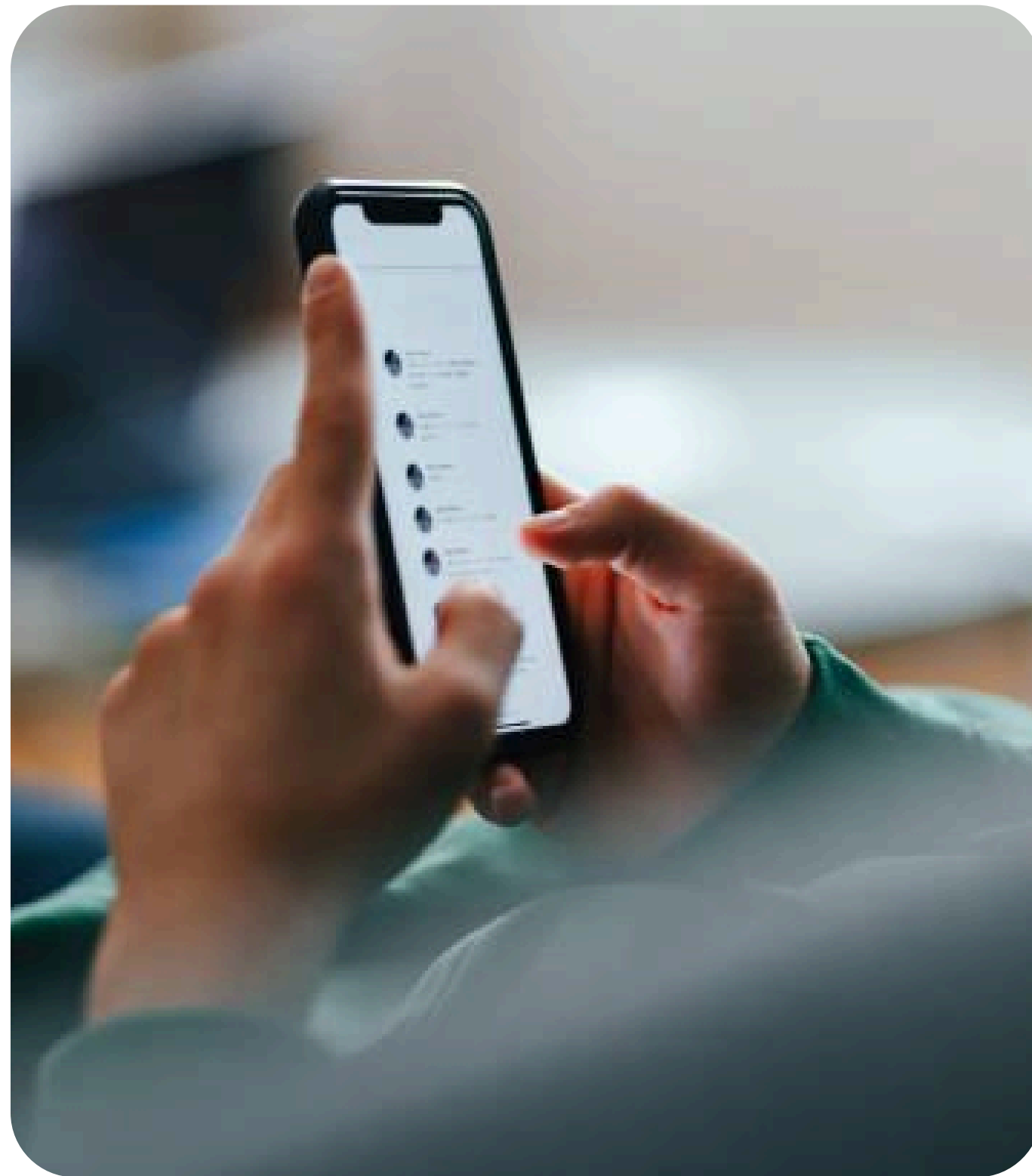
High concurrency

3

Geolocation
failures

Emergency

Emergent Behaviors



WhatsApp



Geographic inequality



Supplier disengagement

Chaotic Dynamics

- 1 Reassignment cascades
- 2 Notification amplification
- 3 Negative metric feedback loops





From System to Algorithm

How system constraints shaped the design

TABLE IV
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REQ-03	Mandatory priority for verified charities on large quantities	Matching algorithm shall apply priority weighting to verified charity accounts	$\geq 90\%$ of offers ≥ 10 kg assigned to charities first	Reliability
REQ-04	Geolocation as critical input and single point of failure	System shall validate geolocation in real time with fallback mechanism	Automatic rejection if invalid; fallback available in < 5 seconds	Reliability
REQ-05	Transport, quality verification and payment outside system boundaries (Section I-B.2)	Platform shall restrict its functionality exclusively to digital coordination, reservation and collection tracking of food surplus	0 modules related to payment processing, physical transport management or food quality verification present in the system architecture (verified by code review and system testing)	Maintainability
REQ-06	High sensitivity to no-shows	System shall implement automatic timeout and re-assignment	Timeout = 15 minutes; re-assignment success rate $\geq 85\%$	Reliability
REQ-07	Risk of informal external coordination networks (Section III-C.2)	System shall monitor anomalous acceptance patterns	Detection and logging of > 3 coordinated acceptances from the same user group within a 30-minute window for the same offer (audit log mandatory)	Security
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17 Traceable Requirement

Every design decision originates from the system analysis. The algorithm was not merely a technical choice.

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Traceability: Examples

Geographic inequality → REQ-11 → 40% weight assigned to verified organizations.

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Traceability: Examples

Notification amplification $\rightarrow$ REQ-12
 $\rightarrow$ tiered notification strategy.

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Traceability: Examples (Cont.)

No-show cascade $\rightarrow$ REQ-08 $\rightarrow$ automatic reassignment with compressed timeout.

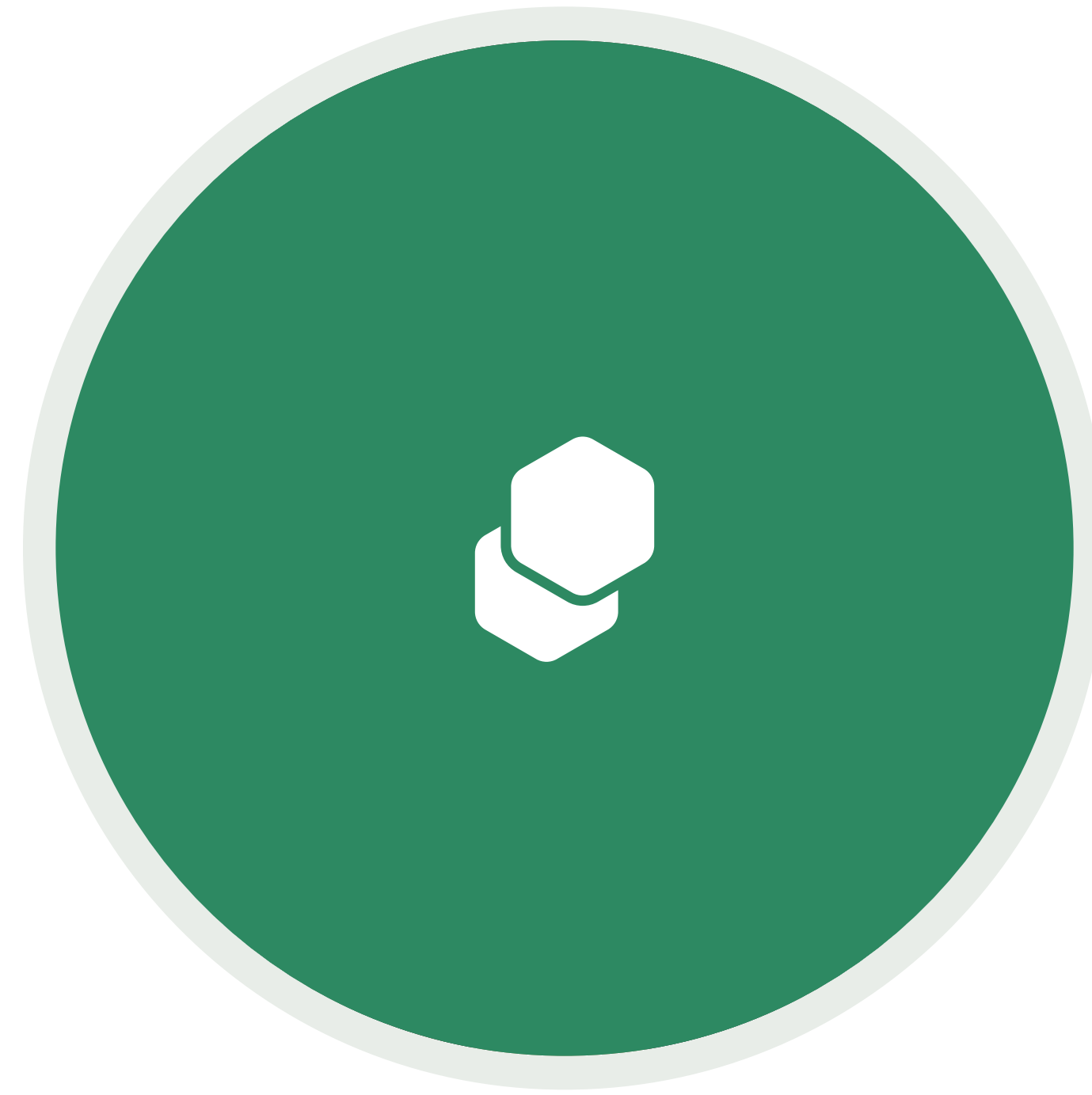
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Traceability: Examples (Cont.)

High concurrency → REQ-09 → priority queue serialization + database locking.



The *matching* formula.

score = 0.4×proximity + 0.4×charity_bonus + 0.2×reliability

Why These Weights?

1 Proximity (0.4)

Ensures logistical feasibility within a 1–3 hour collection window.

3 Charity Bonus (0.4)

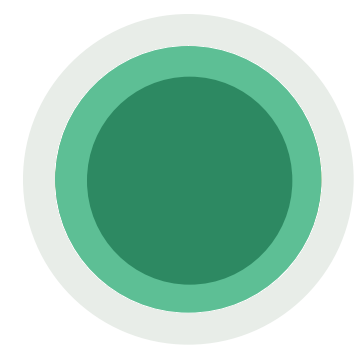
Offsets proximity bias in favor of verified organizations.

3 Reliability (0.2)

Discourages no-shows through asymmetric penalties (+1 / -2).

“

A no-show harms the system more than a successful pickup benefits it. Queue serialization prevents race conditions during high-concurrency scenarios.



Why the Asymmetry Is Not Arbitrary

Donor-to-Beneficiary Flow

- 1 Posting activates real-time matching.
- 2 Tiered notifications: 500 m $\rightarrow$ 1 km $\rightarrow$ 3 km, with delays of 10 and 20 minutes.
- 3 15-minute timeout per attempt. Maximum of 3 reassignments with compressed windows.



Chaotic Dynamics

- 1 Reassignment cascades
- 2 Notification amplification
- 3 Negative metric feedback loops

What This Flow Solves

It directly addresses the three chaotic dynamics identified in Workshop 1: cascades, amplification, and geographic bias.



Evidence: Simulation Proves It

Evidence: Simulation Proves It



Why simulate before building?

Simulating first was not a course requirement. It was the correct engineering decision.

The Two Paradigms

DES (SimPy 4.1)

150 replications, event-driven flow, matching latency analysis.

ABM (Mesa 2.3)

150 replications, 5 agent types, emergent behavior analysis.

Reproducibility conditions

300 total runs with convergent results across paradigms. Deterministic seeds 0–29 ensure full reproducibility.

Table VII: Main Simulation Results (Mean $\pm$ 95% CI, $n = 30$)

Scenario	Recovery Rate	Completion Rate	Charity Share	No-Show Rate	Reassignments
Low Adoption	0.997 $\pm$ 0.004	0.804 $\pm$ 0.028	0.247 $\pm$ 0.050	0.051 $\pm$ 0.011	7.6 $\pm$ 1.4
Medium Adoption	0.999 $\pm$ 0.001	0.824 $\pm$ 0.015	0.249 $\pm$ 0.049	0.045 $\pm$ 0.005	13.0 $\pm$ 1.3
High Adoption	0.998 $\pm$ 0.002	0.819 $\pm$ 0.013	0.337 $\pm$ 0.034	0.054 $\pm$ 0.006	26.5 $\pm$ 2.3
No Mitigations	0.982 $\pm$ 0.007	0.638 $\pm$ 0.015	0.289 $\pm$ 0.045	0.108 $\pm$ 0.008	32.7 $\pm$ 2.0
Sensitivity	0.962 $\pm$ 0.009	0.558 $\pm$ 0.016	0.308 $\pm$ 0.046	0.151 $\pm$ 0.014	44.0 $\pm$ 2.7

Table VIII: Validation Against Project Success Criteria

Scenario	Recovery $\geq$ 80%	Completion $\geq$ 85%	Charity $\geq$ 20%	Overall
Low Adoption	99.7% ✓	80.4% $\triangle$	24.7% ✓	2/3
Medium Adoption	99.9% ✓	82.4% $\triangle$	24.9% ✓	2/3
High Adoption	99.8% ✓	81.9% $\triangle$	33.7% ✓	2/3
No Mitigations	98.2% ✓	63.8% $\times$	28.9% ✓	2/3
Sensitivity	96.2% ✓	55.8% $\times$	30.8% ✓	2/3
✓ = Met $\triangle$ = Near miss (< 3 pp below threshold) $\times$ = Not met				

Table IX: DES Validation Against Project Success Criteria

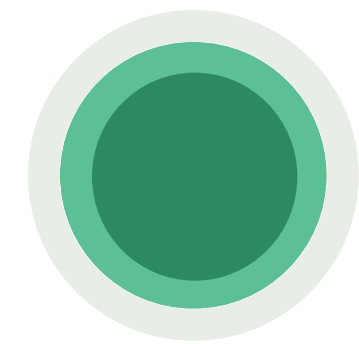
Scenario	Recovery $\geq$ 80%	Completion $\geq$ 85%	Charity $\geq$ 20%	Overall
Baseline	98.6% ✓	89.4% ✓	98.5% ✓	3/3
Early Stage	98.3% ✓	85.2% ✓	97.8% ✓	3/3
Stress Test	98.9% ✓	92.2% ✓	99.7% ✓	3/3
Risk Test	96.8% ✓	78.5% $\times$	99.8% ✓	2/3
Sensitivity	95.7% ✓	70.3% $\times$	99.8% ✓	2/3
✓ = Met $\times$ = Not met				

Key Results

- Recovery rate: 96–99% across all scenarios.
- Pickup completion rate: 89–92% under nominal conditions.
- Matching engine utilization: <0.4% — capable of scaling 10–20× without modification.

“

The system has substantial excess capacity. User behavior is the dominant risk factor.



Simulation Conclusion

Results

2 out of 3 Criteria Achieved

$\geq 80\%$

Recovery Rate;
achieved in all 10
scenarios

$\geq 20\%$

Charity
Assignment
Share

$\geq 85\%$

Pickup
Completion

82.4% — a near miss of 2.6 percentage points.

Why the Gap Is Not a Failure?

The model uses fixed no-show probabilities, which do not capture the adaptive learning behavior of real users penalized by the reliability system.

The solution is not to modify the algorithm.

Closing the Gap

1 Confirmation

Add explicit confirmation before assignment.

2 Commitments

Establish institutional commitments with verified organizations

3 System Vs Software

These are system-level solutions, not software solutions.



The Application as a Direct Solution

Technology serving the system



Live Demo

📁 Catch-Up	Add: Slides	3 months ago
📁 Workshop_1_Analysis	Update README.md Workshop 1, Issue #56	2 months ago
📁 Workshop_2_Design	Update README.md Workshop 2, Issue #56	2 months ago
📁 Workshop_3_Management	Update README.md	2 months ago
📁 Workshop_4_Simulation	Merge pull request #107 from Cristian7B/feature/pos	4 days ago
📁 Workspace	fix: Readme layout & add: All docs for the workshop	3 months ago
📄 Paper~ Food Waste Reduction Platform.pdf	Add files via upload	1 hour ago
📄 Poster~ Food Waste Reduction Platform.png	Add files via upload	1 hour ago
📄 README.md	Update README.md	5 days ago

Full Traceability: W1 → W2 → W3 → W4 → App

Every identified risk has a corresponding mitigation strategy that was designed, validated through simulation, and implemented in the application.

Risk → Requirement → Implementation (I)

1 R1 (race conditions)

Serialization + DB locking → NestJS + PostgreSQL transactions.

2 R4 (no-shows)

Reliability scoring + timeout → Visible score in the UI + automatic reassignment.



Risk → Requirement → Implementation (II)

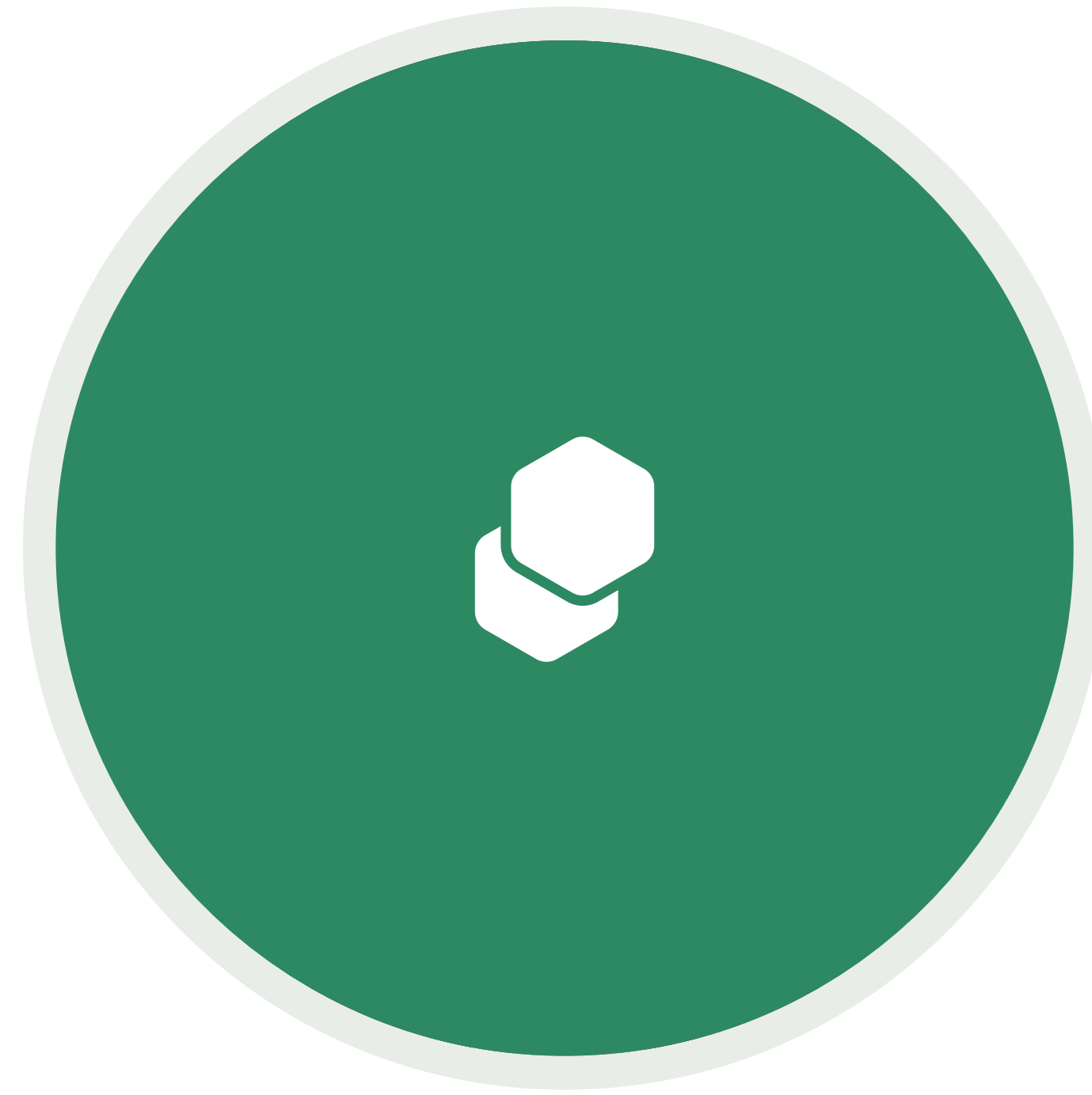
1 R7 (inequity)

40% fairness weight → Equity index displayed in the dashboard.

2 R9 (amplification)

Tiered notifications → Firebase tiers at 500 m / 1 km / 3 km.





The *key* Architectural Insight.

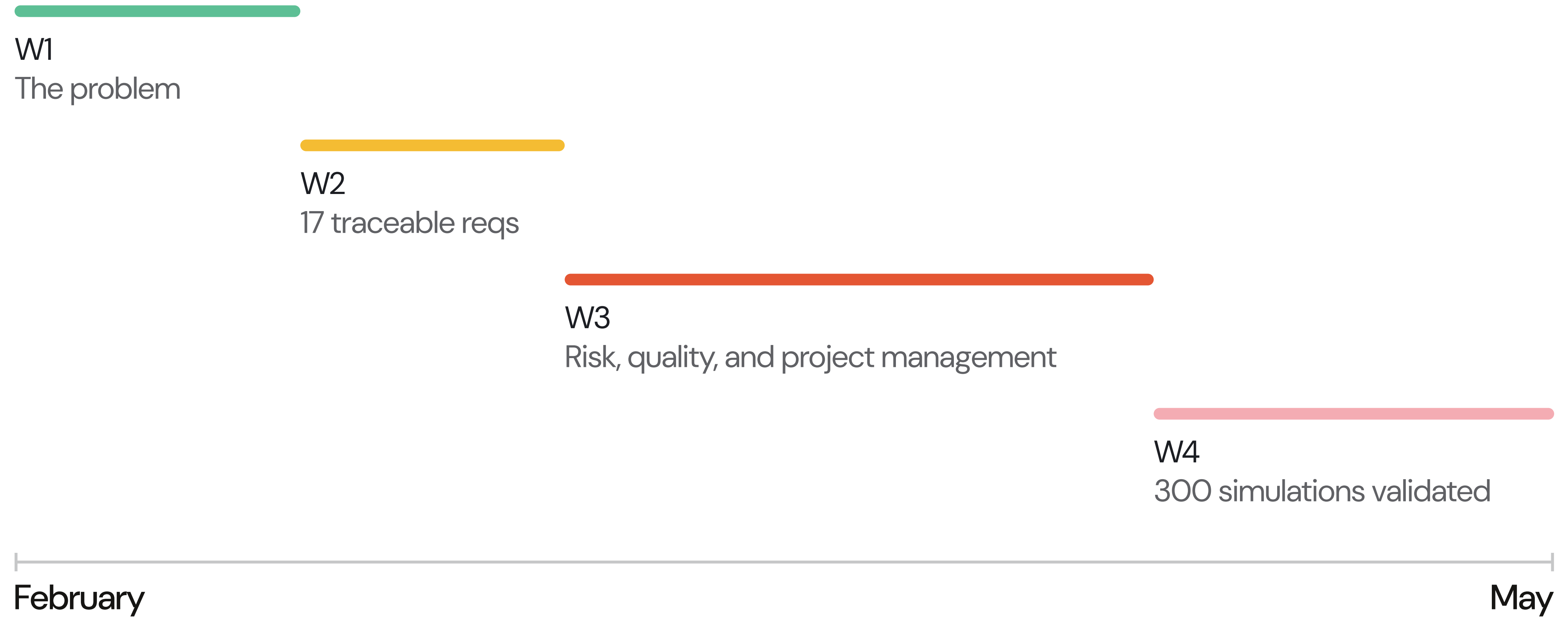
There is not a single technical decision in the application without traceability.



What We Learned from the System

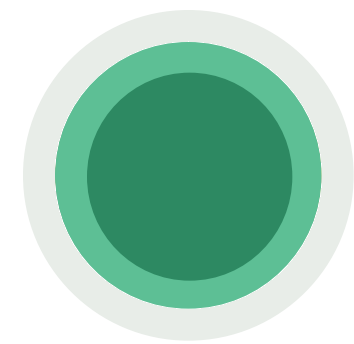
Closing Remarks

Four Workshops, One Common Thread



“

Technology can optimize coordination, but it cannot force human behavior.



The Central Finding



Closing

The application is not the project.
It is the evidence that the analysis was correct.

Questions and Discussion